Department Overview

- **Department Name:** Department of Mathematical Sciences
- **Faculty:** Faculty of Basic Sciences
- **History:** Established in 2003.
- **Core Focus:** Encompasses pure, applied, computational, and financial mathematics. Research interests span from abstract mathematical theory to practical, applied aspects of the discipline.
- **Role & Instructional Support:** Provides baseline instructional support across all university faculties for courses requiring mathematical sciences.
- **Student Demographics:** Welcomes students hailing from Balochistan in particular, other parts of Pakistan in general, and foreign students on limited reserved seats.

Vision and Goals

- **Vision:** To be among the leading Mathematics departments of the country, which provides quality education in Mathematics and is the centre of active and innovative research.
- **Core Philosophy:** To promote the understanding of mathematics through teaching and research, keeping pace with the latest global trends while inculcating logical and critical thinking attributes in its students.

1. Bachelor of Science in Mathematics (BS Mathematics)

Admission Requirements

- **Academic Background:** Intermediate with Mathematics from any recognized board (or equivalent qualification) with at least **45% marks**.
- **Testing:** Candidates must clear the entrance test conducted by NTS.

Degree Requirements

- **Total Credit Hours:** 128
- **Total Courses:** 42
- **Minimum Academic Standing:** CGPA $\ge$ 2.0

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote (Theory + Lab).

- **1st Semester:** Calculus I (4+0), Elements of Set Theory and Mathematical Logic (3+0), Islamic Education/Ethics (2+0), Introduction to Computers (2+1), English I (Functional English) (3+0), Environmental Sciences (3+0). **Total: 18 CH**
- **2nd Semester:** Calculus II (3+0), Software Packages (Introduction to MATLAB or Maple) (1+2), Introduction to Statistics (3+0), English II (Communication Skills) (3+0), Pakistan

Studies (2+0), Introduction to Sociology (3+0). **Total: 17 CH**
- **3rd Semester:** Algebra I (Group Theory) (3+0), Calculus III (4+0), English III (Technical Report Writing and Presentation Skills) (3+0), Linear Algebra (3+0), Introduction to Psychology (3+0). **Total: 16 CH**
- **4th Semester:** Affine and Euclidean Geometry (3+0), Computational Linear Algebra (1+2), Discrete Mathematics (3+0), Numerical Methods (3+1), Principles of Economics (3+0). **Total: 16 CH**
- **5th Semester:** Point Set Topology (3+0), Classical Mechanics (3+0), Ordinary Differential Equations (3+0), Real Analysis I (3+0), Algebra II (Rings and Fields) (3+0). **Total: 15 CH**
- **6th Semester:** Differential Geometry (3+0), Partial Differential Equations (3+0), Numerical Analysis (3+1), Complex Analysis (3+0), Real Analysis II (3+0). **Total: 16 CH**
- **7th Semester:** Number Theory (3+0), Functional Analysis (3+0), Elective-1 (3+0), Elective-2 (3+0), Mathematical Methods (3+0). **Total: 15 CH**
- **8th Semester:** Probability Theory (3+0), Integral Equations (3+0), Elective-3 (3+0), Elective-4 (3+0), Project (3+0). **Total: 15 CH**

Elective Courses Options

Mathematical Statistics I & II, Fluid Dynamics I & II, Electricity and Magnetism I & II, Theory of Relativity I & II, Quantum Mechanics I & II, Algebraic Topology, Approximation Theory, Fixed Point Theory, Measure Theory I & II, Theory of Modules, Advanced Group Theory, Graph Theory, Algebraic Coding Theory, Representation Theory of Finite Groups, Elliptic Curves, and Seismology.

2. Master of Science in Mathematics (MS Mathematics)

Admission Requirements

- **Academic Background:** 16 years of education in Mathematics from an HEC-recognized institution.
- **Testing:** A minimum score of **50%** on the GAT (General) test conducted by NTS.

Degree Requirements

- **Total Credit Hours:** 30
- **Total Courses Listed:** 32 *(Note: The raw web copy contains a text layout error erringly listing "Total Courses: 32" for a standard 30 credit hour MS track).*
- **Minimum Academic Standing:** CGPA $\ge$ 2.5

Program Schema Structure

- **1st Semester:** Core-1 (3+0), Core-2 (3+0), Core-3 (3+0), Core-4 (3+0). **Total: 12 CH**
- **2nd Semester:** THESIS-601 Thesis. **Total: 6 CH**

Available Graduate Course Options

- **Core Courses Pool:** Theory of Group Actions, Riemannian Geometry, Algebraic Topology, ODEs and Computational Linear Algebra, Integral Equations, Theory of Partial Differential Equations, Numerical Solutions of PDEs, Advanced Mathematical Physics, Special Functions, Research Methodology, Rings and Modules, General Theory of Relativity, Near Rings-I, Advanced Ring Theory-I, Commutative Algebra-I, Homological Algebra-I, Magnetohydrodynamics-I, Advanced Analytical Dynamics-I, Plasma Theory-I, Multivariable Analysis-I, and Algebraic Number Theory.
- **Elective Courses Pool:** Differentiable Manifolds, History of Mathematics, Approximation Theory, Fixed Point Theory, Lattice Theory, Commutative Algebra, Category Theory, Axiomatic Set Theory, Theory of Ordinary Differential Equations, Classical Electrodynamics-I & II, Representation Theory-I & II, Computer Aided Geometric Design (CAGD), Multi-resolution Geometric Modeling, Mathematical Modeling, Perturbation Methods I, Geometric Function Theory, Fluid Mechanics, Banach Algebras, Theory of Group Graphs, Lie Groups and Lie Algebras, Semi Group Theory, LA-Semigroups, Near Rings-II, Several Complex Variables, Non-Standard Analysis, Advanced Ring Theory-II, Topics in Complex Analysis, Loop Groups, Nilpotent and Soluble Groups, Commutative Algebra-II, Commutative Semigroup Rings, Homological Algebra-II, Theory of Semigroups, Fuzzy Algebra, Algebraic Coding Theory, Magnetohydrodynamics-II, Introduction to Algebraic Cryptography, Advanced Analytical Dynamics-II, Mathematical Techniques for Boundary Value Problems, Plasma Theory-II, Astrophysics, Non-Newtonian Fluid Mechanics, Multivariable Analysis-II, Statistical Mechanics, Linear Systems Theory, and Heat Transfer.

3. Doctor of Philosophy in Mathematics (PhD Mathematics)

Admission Requirements

- **Academic Background:** An HEC-recognized MS or equivalent 18-year master's degree. The candidate's degree must feature a minimum of **30 credit hours** (consisting of 24 credit hours of graduate-level coursework + 6 credit hours of research thesis).
  - *Note: The source text page layouts errantly retain a structural typo copy-pasted from another branch, erringly naming the degree domain as "Chemistry" within this criteria block.*
- **Academic Performance:** Minimum of a 1st Division pass under the annual system, or an equivalent program CGPA of **3.0 out of 4.0**.
- **Standardized Testing:** Prior to confirmation of program admission, candidates must successfully pass either the standard **GRE (International) Subject Test** with a minimum 60th percentile score, or an authorized **GAT Subject Test** with at least **60% marks**.

Degree Requirements

- **Total Credit Hours:** 18
- **Total Courses:** 6 advanced doctoral courses.
- **Minimum Academic Standing:** CGPA $\ge$ 3.0